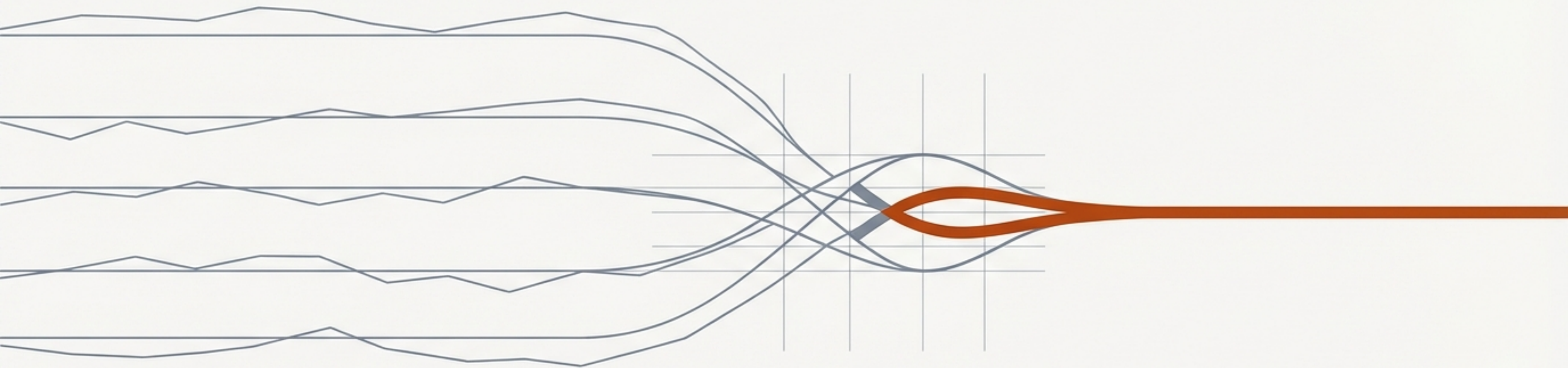


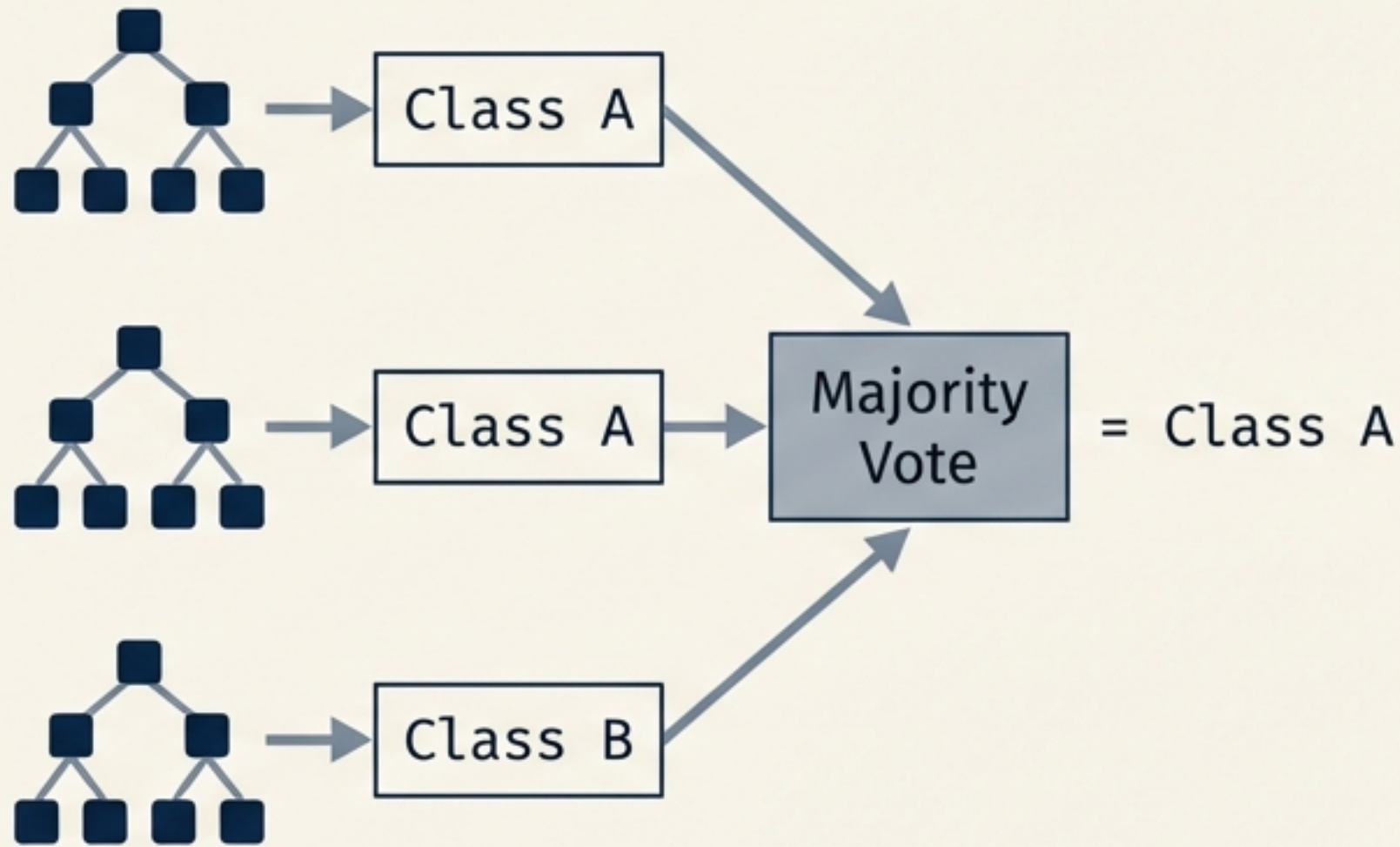
An implementation guide for scikit-learn.



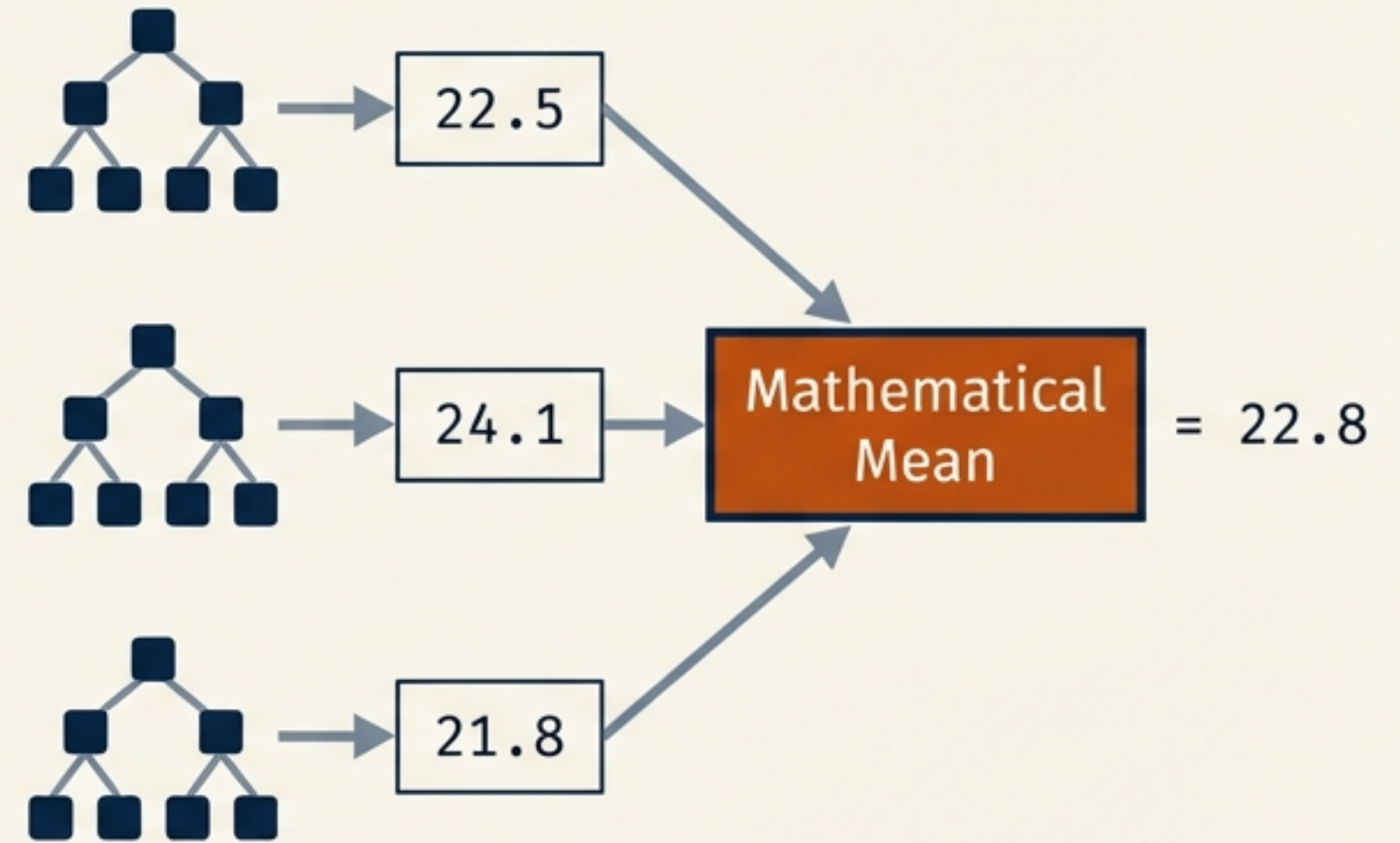
Mastering the Bagging Regressor

Adapting Bagging for Continuous Variables

Classification

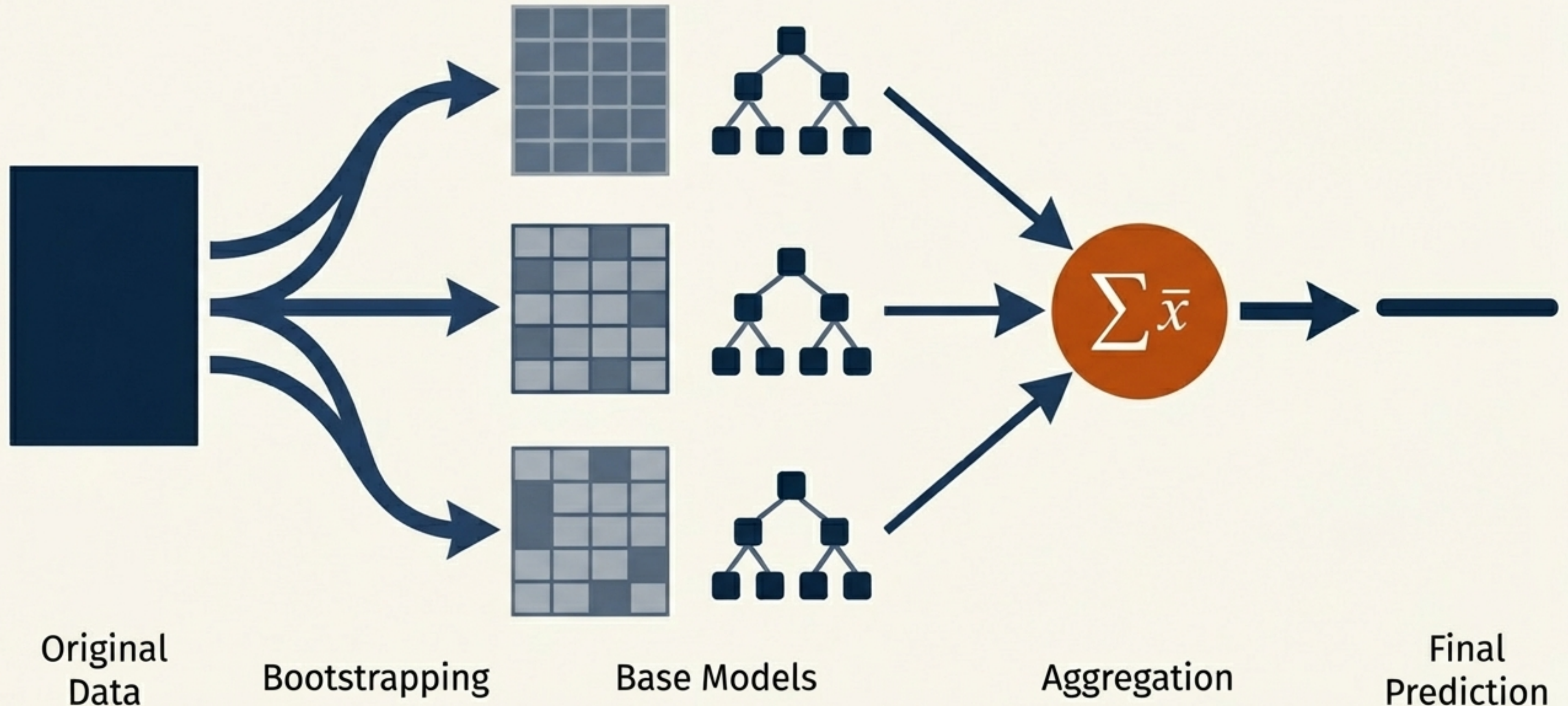


Regression



The core algorithm remains identical. The only shift occurs at the final aggregation step: we replace majority voting with the mean.

The Bootstrapping and Aggregation Pipeline

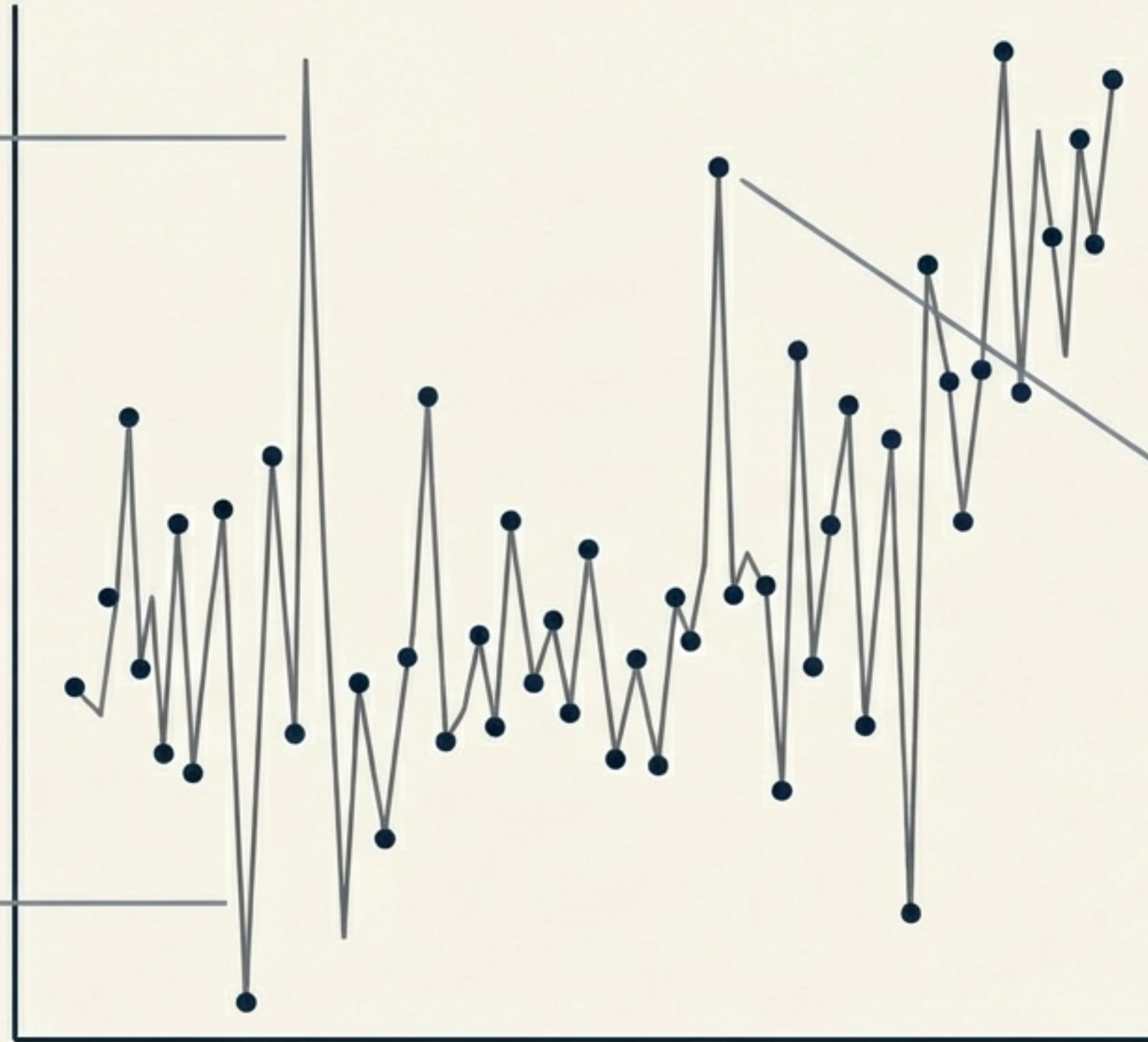


The Problem with Single Trees

Low Bias,
High Variance.

The model avoids
errors on training
data but fails to
generalise.

Overfitting: Capturing
noise instead of the
underlying trend.



Variance Reduction through Ensembling



The Testing Arena: Boston Housing



Dataset:	Boston Housing
Features (Inputs):	13 numeric columns
Target (Output):	House Prices
Total Records:	506 rows
Data Split:	404 Training points / 102 Testing points

Establishing the Baselines

1	Linear Regression	R2: 0.75
2	K-Nearest Neighbours	R2: 0.54
3	Decision Tree	R2: 0.43

Consultant Note: A single Decision Tree performs poorly on this dataset. It needs reinforcement.

Enter the Vanilla Bagging Regressor

1

Vanilla Bagging Regressor

R2: 0.81

2

Linear Regression

R2: 0.75

Using just the default settings (10 estimators), Bagging instantly outperforms the best single base model.

Exploring the Hyperparameter Space



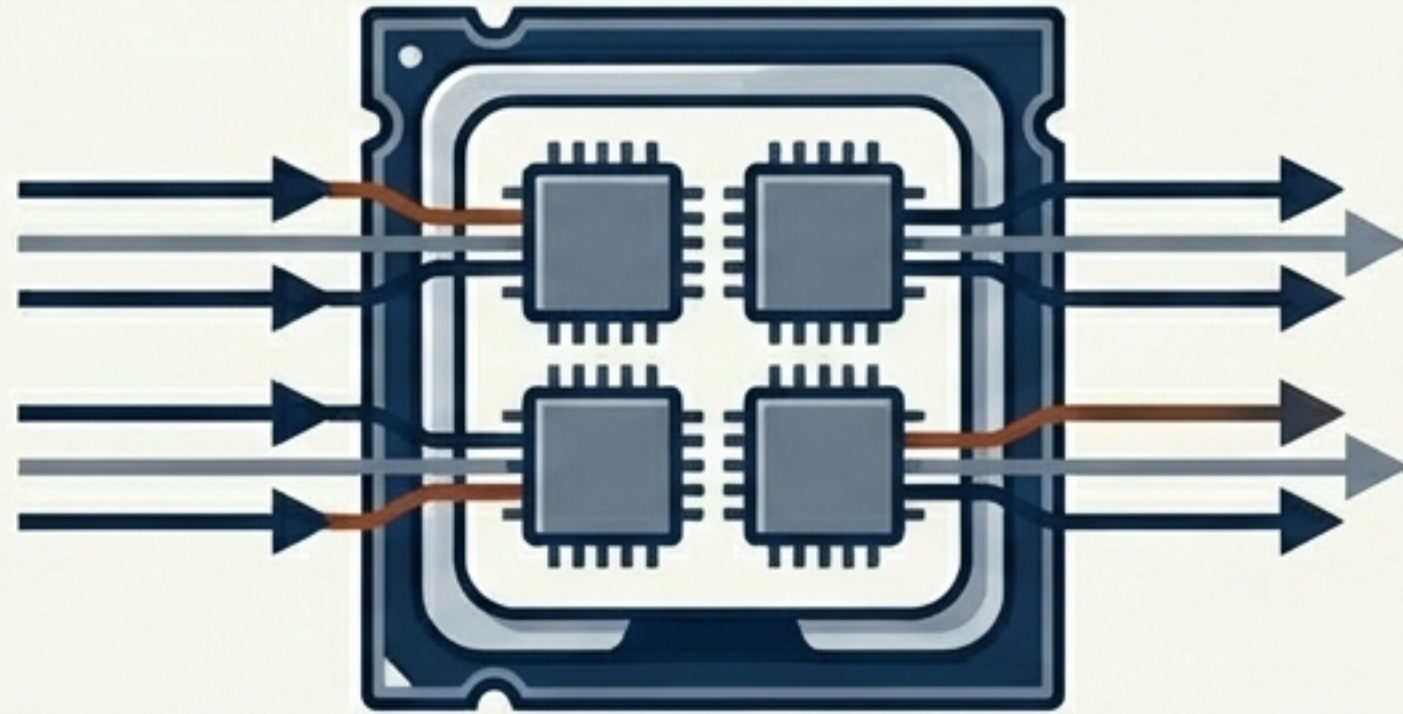
Structuring the Grid Search

```
base_estimator: [Decision Tree, Linear Regression, SVR]
n_estimators: [20, 50, 100]
max_samples: [0.5, 1.0]
max_features: [0.5, 1.0]
bootstrap: [True, False]
bootstrap_features: [True, False]
```

Testing every combination to squeeze maximum predictive power from the dataset.

Accelerating the Search

n_jobs = -1



Grid Search is computationally heavy. Setting `n_jobs=-1` forces the process to utilise all available CPU cores, enabling parallel processing and drastically reducing execution time.

The Winning Configuration

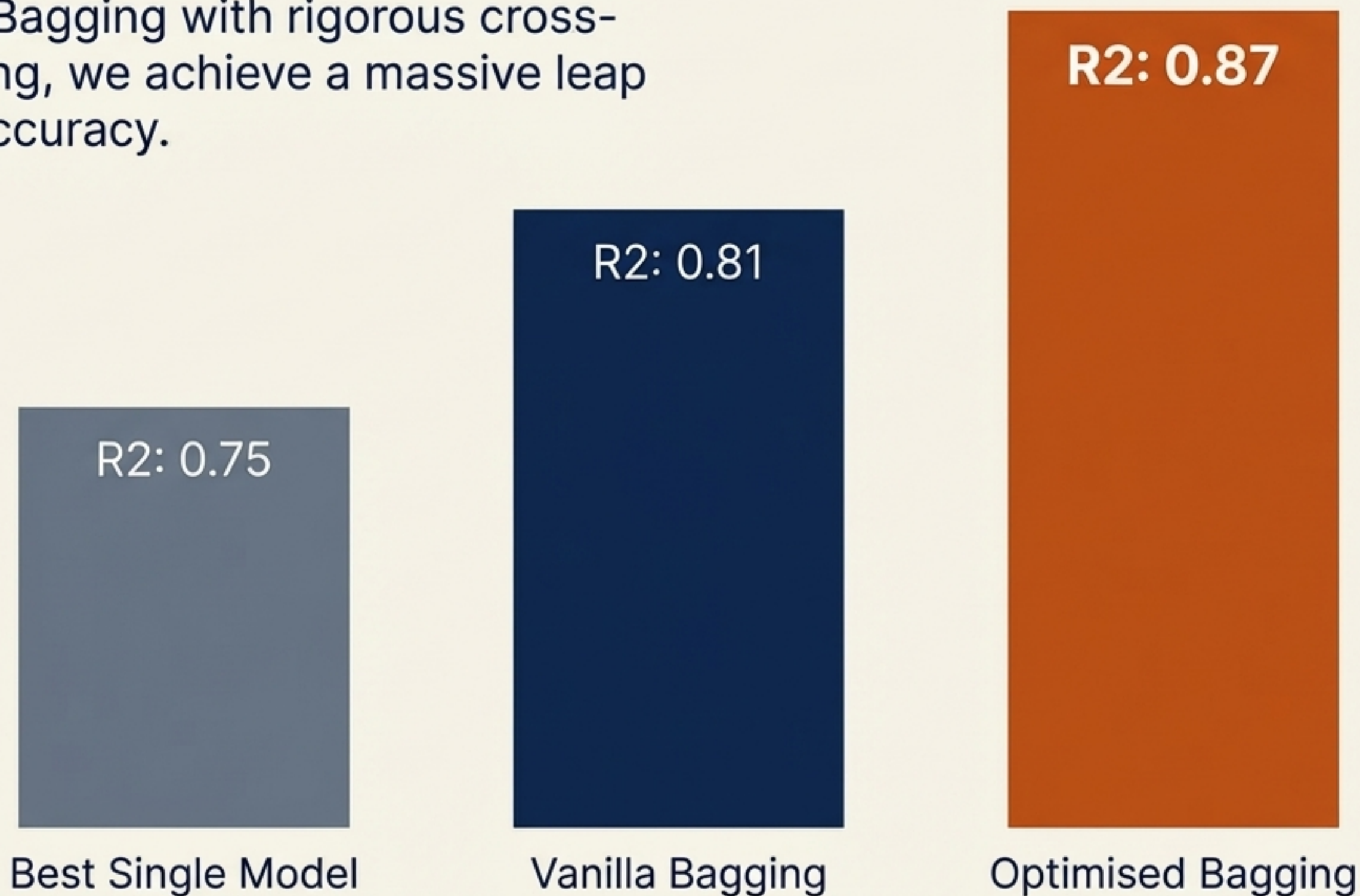


base_estimator: Decision Tree
n_estimators: 50
max_samples: 1.0
max_features: 1.0
bootstrap: True
bootstrap_features: False

The optimal setup relies on 50 Decision Trees using standard row bootstrapping without column sampling.

The Final Benchmark

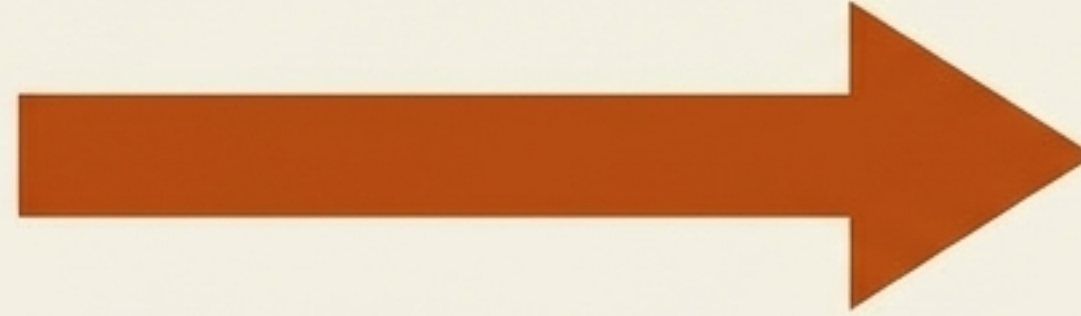
By combining Bagging with rigorous cross-validation tuning, we achieve a massive leap in predictive accuracy.



The Next Evolution



Bagging Regressor



Random Forest

Bagging establishes the foundation of variance reduction. By introducing further feature-level randomness at every node split, we arrive at the ultimate implementation of this technique: The Random Forest.